DEEP LEARNING LABORATORY

EXPERIMENT 11

CNN FOR IMAGE CLASSIFICATION

Aim

To implement a Convolutional Neural Network (CNN) using TensorFlow/Keras for handwritten digit image classification with the MNIST dataset, and evaluate the model using training and validation accuracy.

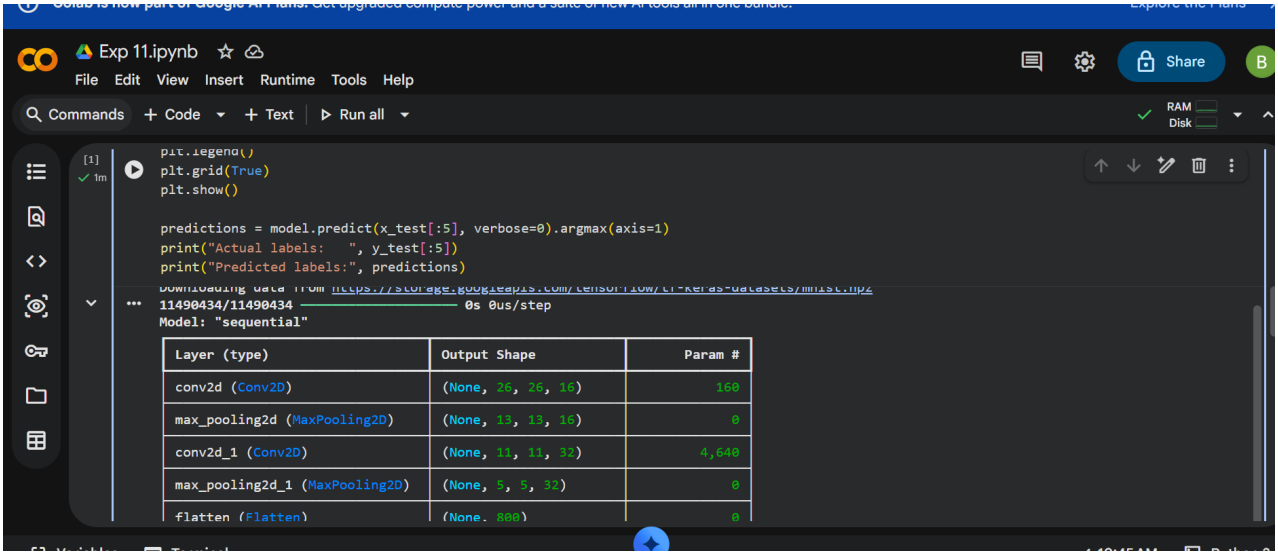
Output

The model summary displays convolutional, max-pooling, flattening, and dense layers. The accuracy graph shows training accuracy increasing across three epochs, while validation accuracy remains high and improves slightly.

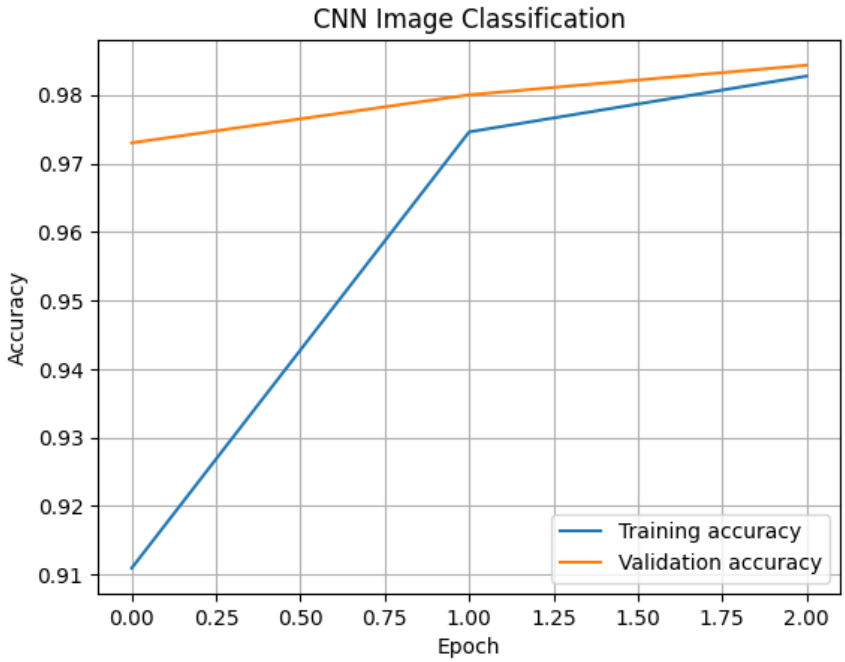
Observed output

Item	Observed output
Dataset	MNIST handwritten digits
CNN layers	Conv2D, MaxPooling2D, Flatten, Dense
Epochs shown	3
Training accuracy	Approximately 98.3% at final epoch
Validation accuracy	Approximately 98.4% at final epoch

Colab output screenshot



Accuracy graph


CNN Image Classification
Accuracy
Epoch
0.98
0.97
0.96
0.95
0.94
0.93
0.92
0.91
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Training accuracy
Validation accuracy